THE UNPLANNED BASKET

How quick-commerce is quietly converting India's planned weekly grocery list into a daily stream of impulse spending.

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Dataset: 13,000 SKUs · 10 cities · 24 months · Rs 509M revenue

The Core Question

Why this project exists — the kirana paused you. The app doesn't.

What share of Blinkit's sales is **planned need** versus **unplanned want** — and is urban India quietly spending more because the ten-minute app has replaced the weekly list with the daily tap?

Kirana era

- Weekly trip to neighbourhood store
- A written list forced deliberation
- Pause between needing and buying

Quick-commerce era

- ~10-minute delivery SLA
- ~2 lakh kirana closed in 2024
- 3–4 daily taps replace one weekly list

What's unmeasured

- SKU-level impulse share inside one operator
- Per-gram premium on small packs
- City-by-city variance

Dataset at a Glance

Blinkit product-level operational data · 13,000 SKUs · 24 months.

13,000
SKUs analysed

25 + 11
raw + engineered columns

10
Indian cities

24
months · Oct-23 → Oct-25

Coverage

- Cities: Mumbai, Delhi, Bengaluru, Chennai, Hyderabad, Kolkata, Pune, Ahmedabad, Jaipur, Lucknow
- Categories (8): Grocery, Dairy, Personal Care, Household, Beverages, Snacks, Bakery, Fruits & Vegetables
- 28 brands · 6 seller types · 4 weight buckets (small / mid / bulk)
- Total revenue in scope: Rs 509,027,130
- Basket rows — Planned 5,117 · Mixed 4,139 · Impulse 3,744

Known issues & fixes

- offer_type NaN for ~50% rows → recoded as 'No Offer' in Notebook 02
- Mixed date formats → parsed to ISO dates
- No duplicate product_ids; arithmetic consistency verified
- Single snapshot per SKU (no transactional log) — temporal trend uses month-of-date_added as proxy
- Impulse is inferred from SKU attributes, not customer behaviour — findings framed as 'signature of impulse'

Methodology — The Impulse Proxy Score

Turning pack size, shelf life, offer and category into one 0–1 fingerprint.

```
impulse_proxy_score = 0.30·(is_small) + 0.25·(is_perishable) + 0.25·(is_impulse_category) + 0.20·(has_offer)

basket_type = Impulse (score ≥ 0.6) · Planned (score ≤ 0.4) · Mixed (otherwise)
```

1 · Engineer

- price_per_100g for comparable unit economics
- weight_bucket: <300g / 300-700g / >700g
- is_perishable: shelf_life ≤ 30 days
- has_offer flag; is_impulse_category
- 11 derived features written to cleaned.csv

2 · Explore (EDA)

- Univariate: price, quantity, impulse_score distributions
- Bivariate: category × basket type, weight × price
- City-level impulse revenue share
- Pareto on impulse revenue
- Correlation heatmap of 7 metrics

3 · Validate (Stats)

- H1 Mann-Whitney: Impulse vs Planned units sold
- H2 ANOVA: offer vs no-offer on units
- H3 Kruskal-Wallis: 10 cities on impulse score
- H4 Welch t-test: small vs other pack price/100g
- H5 Pareto 80/20 on product revenue

EDA Headline — Revenue Split

Planned is the base. Impulse is a measurable 21.6% worth optimising.

The number

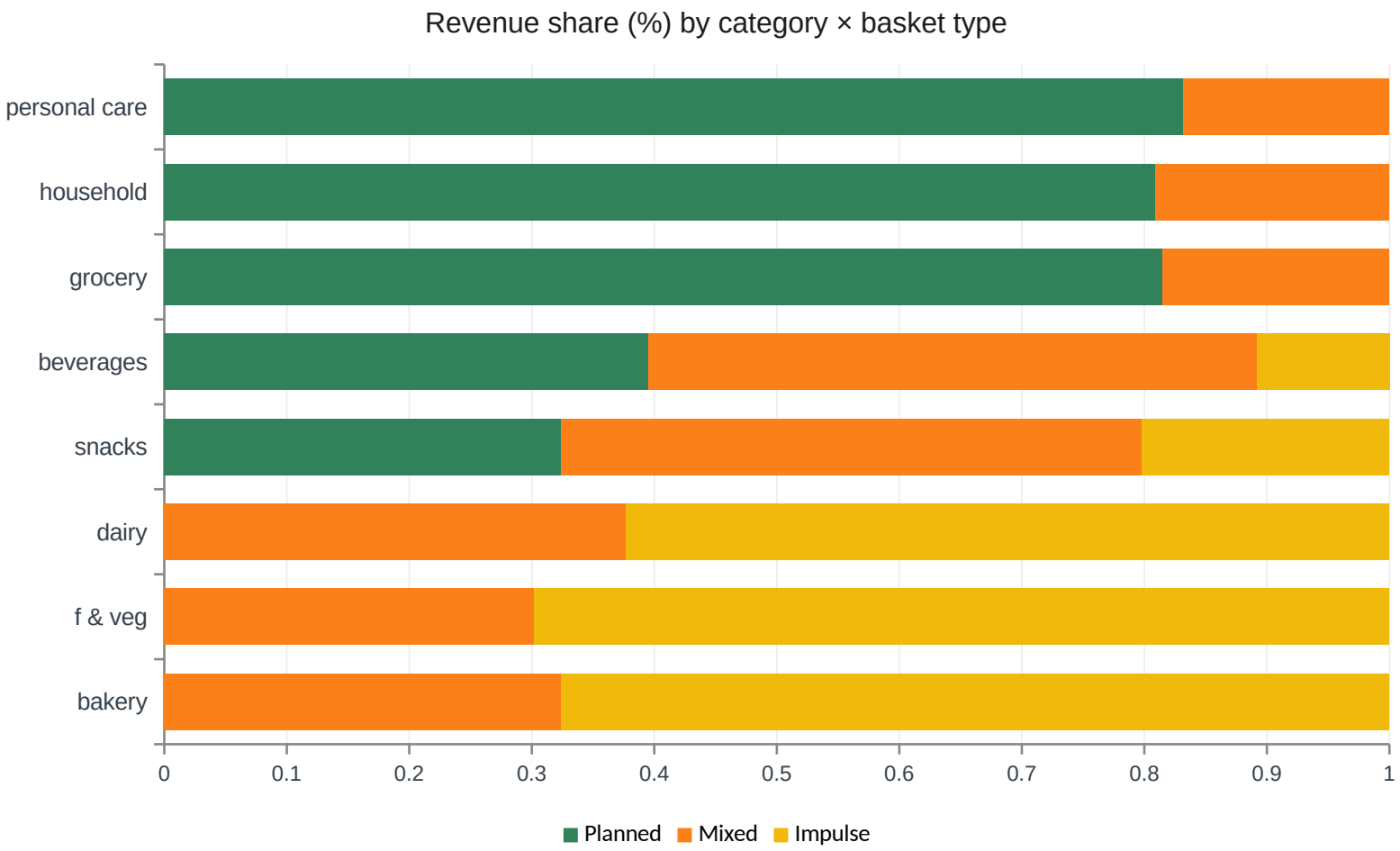
- Total revenue: Rs 509,027,130
- Planned: Rs 260.73M (51.2%)
- Mixed: Rs 138.16M (27.1%)
- Impulse: Rs 110.13M (21.6%)

What this tells us

- Planned dominates revenue; but it only earns 51.2% from 39.4% of SKUs (5,117 / 13,000) — lower SKUs, higher ticket.
- Impulse is 28.8% of rows and 21.6% of revenue — broad behaviour, smaller tickets per SKU.
- Mixed sits between the two — the swing category in merchandising terms.

Where Impulse Lives — By Category

Three categories drive 89% of impulse revenue. Three are pure-planned.



Impulse engines

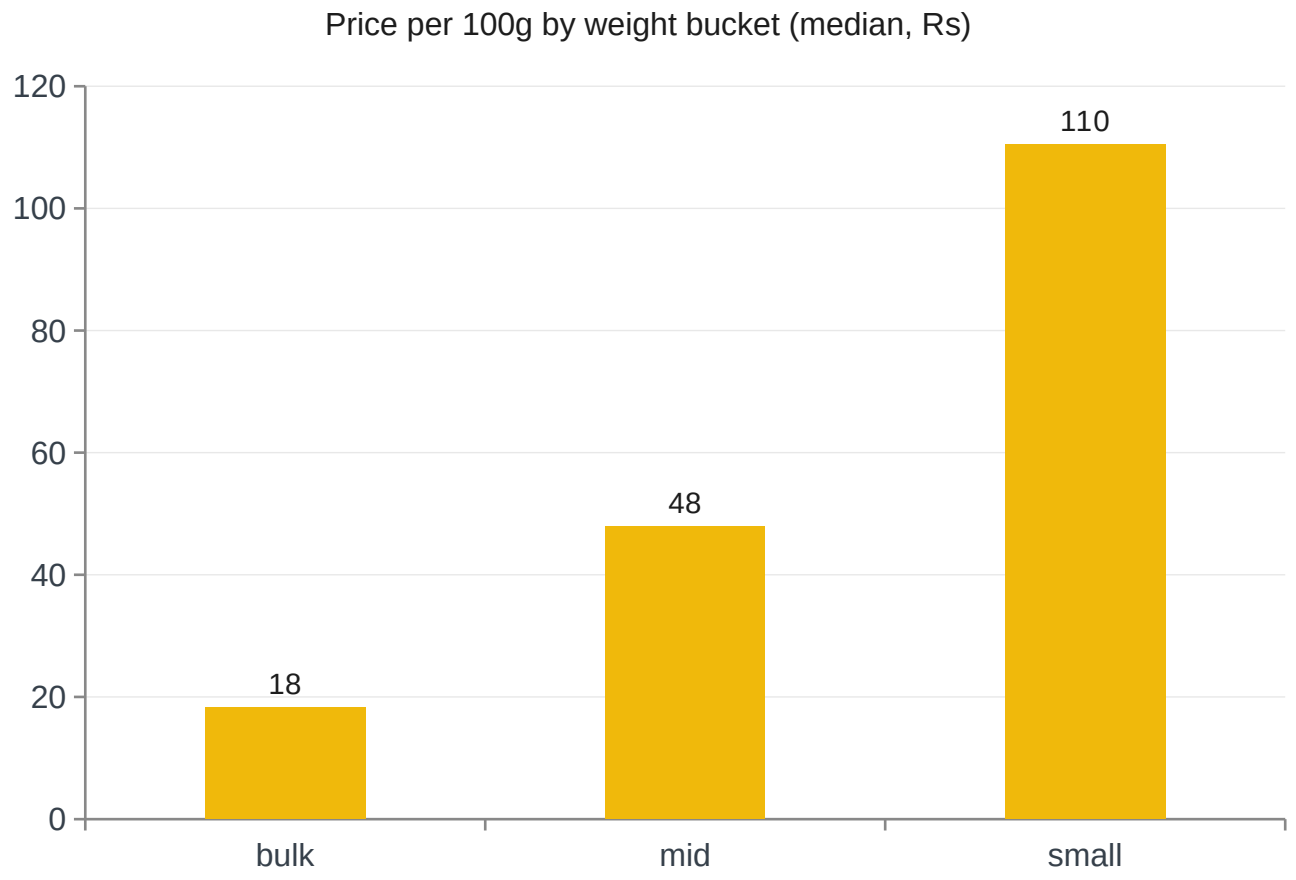
- bakery 67.6% impulse
- f & veg 69.8% impulse
- dairy 62.3% impulse
- Pareto: 89.1% of impulse revenue

Lock-in categories

- personal care 83.2% planned
- grocery 81.5% planned
- household 80.9% planned
- Zero impulse revenue in these three.

Pricing — The Small-Pack Premium

Small packs cost ~6× bulk per 100g. Confirmed by Welch t-test.



6.0×
small vs bulk price/100g

0.80
corr(price/100g, impulse premium)

t = 49.4
Welch t-test

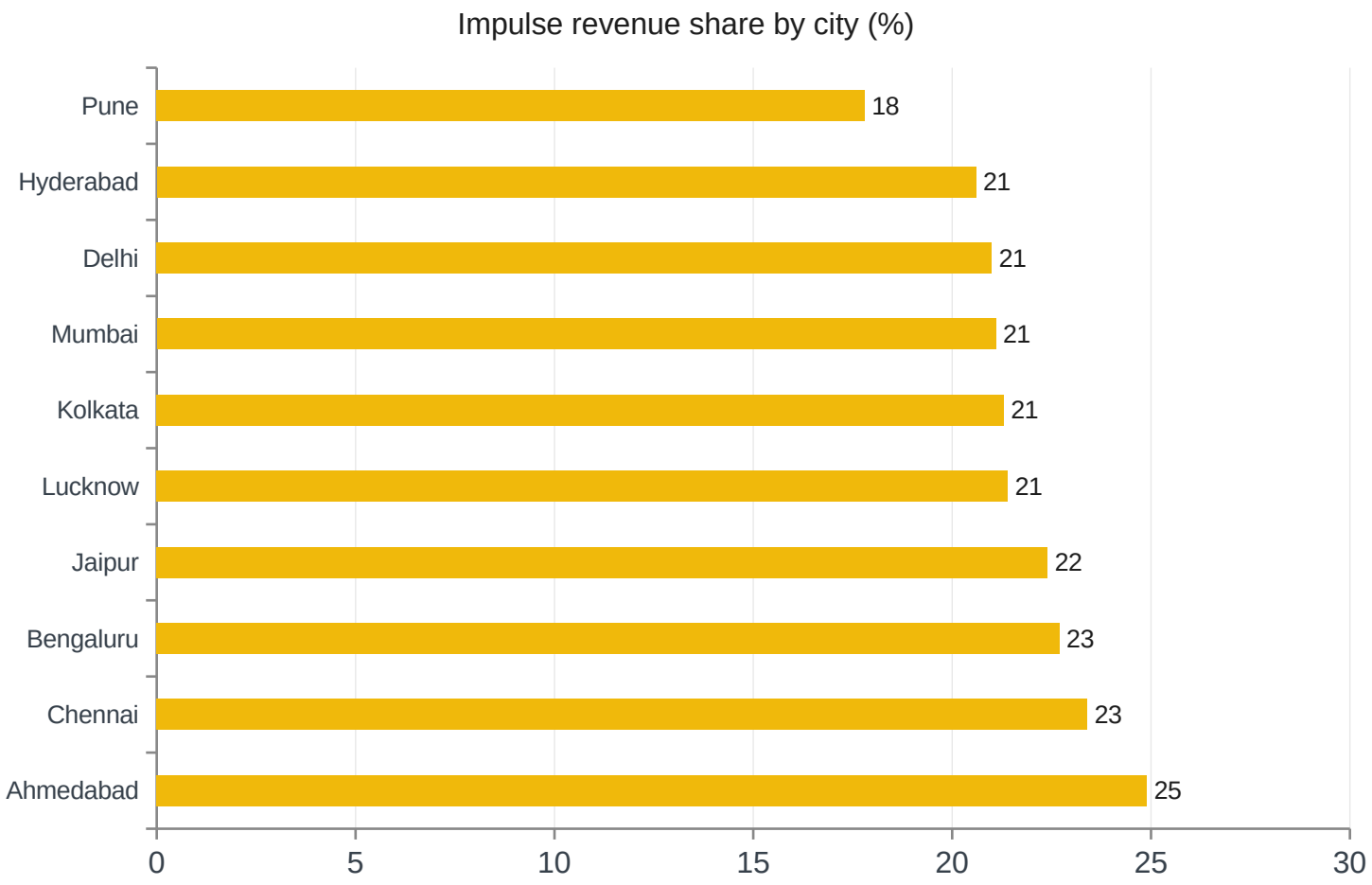
p ≈ 0
small packs cost more (SIG)

Why it matters

- Per-gram price rises sharply as pack shrinks (18 → 48 → 110).
- price_per_100g correlates 0.80 with impulse_premium_pct — higher ppg packs carry a bigger 'tap tax'.
- This is the sharpest, most statistically solid finding in the dataset.

Geography — City Impulse Share

Ahmedabad highest (24.9%), Pune lowest (17.8%). Directional — not statistically significant on behaviour.



The gap

- Ahmedabad 24.9% vs Pune 17.8% = 7.1 pp spread
- Mean across 10 cities: 21.4%
- Ranking is directional for marketing

Caution

- Kruskal-Wallis on impulse_score across cities:
- H = 9.13, p = 0.426 → NOT significant
- Revenue-share gap is real; underlying behavioural score difference across cities is weak.
- Avoid over-reading as 'impulse city vs non-impulse city'.

Statistical Tests — H1 to H5

Only two hypotheses held. The other three are uncomfortable facts for promotions.

H#	Hypothesis	Test	Result	Verdict
H1	Impulse SKUs sell more units than Planned SKUs	Mann-Whitney U	U = 9.6M, p = 0.80	NOT SIG
H2	Offers increase sold_quantity	One-way ANOVA	F = 0.66, p = 0.415	NOT SIG
H3	Cities differ on impulse_score	Kruskal-Wallis	H = 9.13, p = 0.426	NOT SIG
H4	Small packs cost more per 100g	Welch t-test	t = 49.4, p ≈ 0	CONFIRMED
H5	~20% of SKUs drive ~80% of revenue	Cumulative curve	Top 20% = 60.3%	PARTIAL

What actually holds

- H4 — Small packs carry a real per-100g premium (p ≈ 0).
- H5 — Revenue concentration exists, but lighter than 80/20.

What does NOT hold

- Offers / discount depth don't move units in this data (OLS $R^2 \approx 0$).
- Impulse vs Planned median units is effectively equal (124 vs 120).
- Cities are directionally different but not statistically different in behaviour.

Dashboard 1 — Market Pulse

Executive overview — size the prize and the concentration curve.

Q · How big is the impulse opportunity versus the planned base?

Rs 509M
Total revenue

51.2%
Planned · Rs 260.73M

27.1%
Mixed · Rs 138.16M

21.6%
Impulse · Rs 110.13M

DOUGHNUT

Revenue mix by basket type

Confirms the 51 / 27 / 22 split. Planned is the base; Impulse is a broad, material 22% of Rs 509M.

PARETO

SKU Pareto — top 20% → ~60% revenue

Top 20% of products (2,600 SKUs) contribute 60.3% of total revenue — concentration exists but is lighter than classic 80/20.

BAR

Revenue per SKU by basket type

Planned earns more per SKU; Impulse = many SKUs, smaller tickets. Shapes assortment weighting.

CALLOUT

Myth card: Impulse vs Planned units sold

Medians 124 vs 120 units. Mann-Whitney U = 9.6M, p = 0.80. 'Impulse sells in bigger units' is not supported here.

Dashboard 2 — Where Impulse Lives

Category mix — which shelves drive impulse, which are lock-in planned.

Q · Which categories are the engine of impulse, and which are pure planned?

3
impulse-dominant (bakery, F&V, dairy)

3
planned-only (grocery, HH, PC)

2
swing (snacks, beverages)

Rs 42.75M
top category — bakery impulse

100% STACK BAR

Category × basket-type share

bakery 67.6% impulse · F&V 69.8% · dairy 62.3% · grocery 81.5% planned · household 80.9% · personal care 83.2%.

PARETO

Impulse revenue by category

bakery 38.8% cum → F&V 69.0% → dairy 89.1%. Three shelves concentrate 89% of impulse revenue.

TREEMAP

Impulse revenue by category (size)

Visual weight for merchandising priority: bakery 42.75M > F&V 33.22M > dairy 22.17M > snacks 7.46M > beverages 4.54M.

LOLLIPOP

Planned share by category

3 categories clear the 80% planned threshold (personal care 83.2, grocery 81.5, household 80.9) — 'lock-in' set.

Dashboard 3 — Pack Price Ladder

Pricing & pack strategy — where the 'tap tax' shows up in rupees per gram.

Q · Does pack size drive per-gram price, and how do pricing levers relate to demand?

Rs 110
small-pack median Rs/100g

Rs 18
bulk-pack median Rs/100g

6.0×
small vs bulk premium

r = 0.80
price/100g ↔ impulse premium

BOX PLOT

Price per 100g by weight bucket

Medians ladder: bulk 18.32 → mid 47.93 → small 110.44. Welch t = 49.4, p ≈ 0 (H4 confirmed).

SCATTER

Price/100g vs impulse_premium_pct

Correlation r = 0.80 (strong). Higher per-gram priced packs carry a bigger impulse premium.

HEATMAP

7-metric correlation matrix

Signal cells: price ↔ impulse_premium 0.80; qty ↔ demand_index 0.91; qty ↔ revenue 0.64. discount_pct ~0 with everything.

BULLET

Lever check vs sold_quantity

|corr| values: price ~0.00, discount ~0.01, offer ~0.01. Regression R^2 = 0.000. Price/discount/offer are NOT volume levers.

Dashboard 4 — Geography & Promo Myth-Buster

Do cities differ, and do promotions actually move demand?

Q · Where is impulse highest, and do offers or discounts genuinely drive volume?

24.9%
top city — Ahmedabad

17.8%
bottom city — Pune

Δ 1.9
offer vs no-offer mean units

R^2 = 0.000
discount regression

FILLED MAP

Impulse revenue share by city (India)

Yellow ramp on 10 cities. Range 17.8% (Pune) → 24.9% (Ahmedabad). Mean 21.4%. Visual only — behavioural difference is weak.

RANKED BAR

Impulse share ranked by city

Ahmedabad 24.9 > Chennai 23.4 > Bengaluru 22.7 > Jaipur 22.4 > … > Hyderabad 20.6 > Pune 17.8. Gap: 7.1 pp.

BOX PLOT

Sold_quantity by has_offer (0/1)

Means 161.5 vs 163.4 (Δ 1.9). ANOVA F = 0.66, p = 0.415 → offers do NOT lift units. Badge: MYTH.

SCATTER

Discount % vs sold_quantity

Flat bands across 0/5/…/30% buckets. OLS coef 0.15, p = 0.275, R^2 = 0.000. Discount depth fails to move units.

Recommendations for FY26

Three moves Blinkit's Category & Pricing desk can take from this data.

1 · Double down on the three engines

- Bakery + F&V + Dairy = 89.1% of impulse revenue (Rs 97.1M of Rs 110.1M).
- Treat these as the 'impulse shelf': expanded assortment, prominent app placement, stronger small-pack coverage.
- Reduce impulse-style promo spend on grocery / household / personal care — 0% impulse share, planned-demand shelves.

2 · Pack architecture, not discounts

- Small packs carry a 6× per-100g premium (Rs 110 vs Rs 18), confirmed Welch $t = 49.4$, $p \approx 0$.
- Price/100g correlates 0.80 with impulse premium — small-pack expansion is the profitable lever.
- Discount depth shows $R^2 = 0.000$ against units sold — cap discounts on the impulse shelf, redeploy budget to pack design.

3 · Localise with care

- Revenue-share ranking (Ahmedabad 24.9% → Pune 17.8%) is directional only — Kruskal $H = 9.13$, $p = 0.43$.
- Use the ranking for marketing emphasis, not pricing differentiation.
- No evidence yet for 'impulse city' pricing policies in this data — collect customer-level signal before localising prices.

Caveats & Limitations

What the data cannot say — and what would be needed to go further.

Data-side caveats

- Impulse is inferred from SKU attributes, not customer-level behaviour. Findings are the statistical signature of impulse, not causal proof.
- No customer_id → cohort retention, basket size, cross-SKU co-purchase, LTV all out of scope.
- No transactional time-series → no YoY trend, seasonality, or demand-decay claim; temporal uses month-of-date_added as proxy.
- impulse_proxy_score weights (0.30/0.25/0.25/0.20) are analytically chosen — not learned. ±0.1 sensitivity to be reported in notebook 03.
- offer_type was NaN for ~50% of rows; recoded as 'No Offer'. Offer ANOVA therefore compares 'any offer' vs 'unknown/no offer'.
- No Zepto / Instamart data — no competitive benchmark.

Analytical caveats

- H1, H2, H3 are NOT statistically significant — the narrative honours this; avoid 'impulse sells more units' or 'offer lifts demand' claims.
- City ranking (Slide 8 / Dashboard 4) is directional, not a behavioural-difference claim.
- Pareto at SKU level is lighter than 80/20 (top 20% → 60.3%); cite as 'concentrated' not 'classic Pareto'.
- OLS R^2 on sold_quantity using price / offer / discount / impulse_score = 0.000 — those four variables together do not predict volume; demand_index (corr 0.91) does.
- Map in Dashboard 4 uses city aggregate rows; a city drill on behavioural score is not statistically defensible.